International Journal of Heat and Mass Transfer 119 (2018) 10–19

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/ijhmt

The Marangoni effect on microstructure properties and morphology of laser-treated Al-Fe alloy with single track by FEM: Varying the laser beam velocity

Moises Meza Pariona *, André Felipe Taques, Luciano Allan Woiciechowski

Graduate Program in Engineering and Materials Science, State University of Ponta Grossa (UEPG), Ponta Grossa 84010-919, PR, Brazil

ARTICLE INFO

Article history:
Received 11 August 2017
Received in revised form 17 November 2017
Accepted 17 November 2017
Available online 22 November 2017

Keywords:
Laser-treated Al-Fe alloy
Single track
Marangoni effect
Microstructure properties
Morphology
Laser beam velocity

ABSTRACT

Laser workpiece surface treatment of Al-1.5 wt% Fe alloy with single track was performed with a 2 kW Yb-fiber laser (IPG YLR-2000S). A 3D thermal FEM model has been built to calculate the temperature field and fluid flow field in the molten pool cross-section. Results of numerical simulation, experimental of the micrographs and morphologies were presented, discussed and checked. By FEM the fluid flow due to the Marangoni effect presents two symmetric vortices corresponding to the molten pool center line. The Marangoni effect can lead to mass transfer and interfacial turbulence, when this phenomenon is considered, maximum values of the flow velocity and as well as thermal gradient occur around on the workpiece surface, smaller velocity and thermal gradient occur in the melt pool-substrate interface, thus, the Marangoni effect was more noticeable in lower than in high laser beam velocities. This phenomenon influences and controls the quality, such as, properties of the microstructure, morphological characteristic and as well as quality of laser-treated workpiece tracks, therefore, at low laser beam velocities the morphology is higher and quality of track presents many defects than at high laser beam velocities. Simulation results were assessed with the experimental data, their results were fairly coherent.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

Historically several authors have discussed the Marangoni effect. Between them, Siwek [1] argued that the model takes into consideration thermophysical and metallurgical properties of the remelting steel, laser beam parameters and boundary conditions of the process. As a result of heating the material, in the area of laser beam operation a weld pool was created, whose shape and size depends on convection caused by the Marangoni force. Moreover, the Marangoni effect is a frequently observed phenomenon of enhancement of interphase mass transfer in liquid–liquid systems. Such effect, originating from the hydrodynamic instability induced by surface tension sensitivity and surface concentration of transferred solute, which was discussed by the authors Bergman and Keller [2]. In addition, they stated that, the local variation in solute concentration at the interface in liquid–liquid solvent extraction systems would cause local increase or decrease of interfacial tension, and thus induce additional convection at the interface (so-called interfacial turbulence). According to authors Mao and Chen [3], the Marangoni effect is a phenomenon of interfacial turbulence provoked by surface tension gradients that might be induced by gradients in temperature, concentration and surface charge through the interface, still, it is known that this effect effectively enhances the rate of mass transfer across the interface.

According to Gan et al. [4], they affirmed that numerical simulation has previously offered a tool of effective evaluating the thermal behavior during the direct laser deposition process and numerical simulation has offered an effective tool in prediction the thermal behavior and mass transport in direct laser deposition. However, numerical simulation, in general, is a powerful tool to obtain complete understanding of the physical phenomena and the underlying mechanisms of welding processes, as emphasized by Murphy [5]. Yang et al. [6] investigated numerical simulation of heat transfer and fluid flow during laser beam welding, according to authors, the calculated weld pool geometry basically had in good agreement with the experimental results, therefore, these same authors affirmed, to better use this innovative technology, a thorough physical understanding of the associated molten pool phenomena is critically important.

Purpose of this work was to study behavior of the temperature field and fluid-velocity field in the molten pool cross-section involving the Marangoni effect. Results of numerical simulation and microstructure were presented, discussed and checked. More-

* Corresponding author.
E-mail address: mmpariona@uepg.br (M.M. Pariona).

https://doi.org/10.1016/j.ijheatmasstransfer.2017.11.097
0017-9310/© 2017 Elsevier Ltd. All rights reserved.

over, result of influence of the Marangoni effect were shown on surface morphology through SEM and atomic force microscopy (AFM).

2. Mathematical model

A mathematical model involving heat transfer, fluid flow and mass transfer in laser surface remelting (LSR) was proposed. Simplifying assumptions are the following:

1. The fluid flow in the molten pool is assumed to be Newtonian, laminar and incompressible.
2. The thermophysical properties of Al-1.5 wt% Fe alloy were assumed to be temperature-dependent.
3. The laser heat flux is assumed to be a Gaussian distribution.
4. There is no diffusion transport in solid phase.

2.1. Governing equations

Equations of conservation of mass, momentum-transport and thermal energy are formulated in Eqs. (1)–(4). These Equations were proposed by Gao et al. [7] and Gan et al. [4].

Continuity equation,

$$\rho\nabla\cdot(u)=0 \tag{1}$$

where u is the velocity vector of fluid flow in weld pool.

Momentum equation,

$$\rho\frac{\partial u}{\partial t}+\rho(u\cdot\nabla)u=\nabla\cdot[-pI+\mu(\nabla u+(\nabla u)^T]+F \tag{2}$$

where ρ is density, u is the velocity, p is the pressure, I is the identity matrix, T is the temperature and F is the force.

Energy equation,

$$\rho\left(\frac{\partial h}{\partial t}+V\cdot\nabla h\right)=\nabla\cdot(k\nabla T) \tag{3}$$

where h is the enthalpy of material, k is the thermal conductivity.

Equation of heat transfer in fluids:

$$\rho C_p\frac{\partial T}{\partial t}+\rho C_p u\cdot\nabla T+\nabla\cdot q=Q \tag{4}$$

$$q=-k\nabla T$$

where ρ is the density, C_p is the heat capacity at constant pressure, u is the velocity, k is the thermal conductivity, q is the heat flux and Q is the heat.

The Marangoni Effect, Comsol [8]

$$\left[-pI+\mu(\nabla u+(\nabla u)^T)-\frac{2}{3}\mu(\nabla\cdot u)I\right]n=\gamma\nabla T \tag{5}$$

where p is the pressure, I is the identity matrix, μ is the coefficient of viscosity, ∇u is the gradiente of the velocity, T is the temperature, γ is the temperature derivative of the surface tension and ∇T is the gradient of the temperature.

Non-isothermal flow, Comsol [8]

$$-n\cdot q=\rho C_p C_\mu^{\frac{1}{4}}k^{1/2}\frac{T_w-T}{T_{amb}} \tag{6}$$

where n is the coordinate normal to the wall, q is the heat flux, ρ is the density, C_p is the heat capacity at constant pressure, k is the thermal conductivity, T_w is the reference temperature of the wall, T_{amb} is ambient temperature and T is the temperature.

2.2. Boundary and initial conditions

The laser heat flux input at liquid/laser beam interface is as follow, Pariona et al. [9]

$$\rho\frac{\partial(C_PT)}{\partial t}=\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right)+(1-rf)I_0\times\exp\left(-\frac{x^2+y^2}{a^2}\right)exp(-\delta z) \tag{7}$$

On the free liquid surface, Yang et al. [6]:

On the bottom of the workpiece:

$$k\frac{\partial T}{n}=-h_d(T-T_{ref}) \tag{8}$$

On the other surfaces:

$$k\frac{\partial T}{n}=-h_c(T-T_{ref})-\varepsilon k_b(T^4-T_{ref}^4) \tag{9}$$

where h_c is the convective heat transfer coefficient, h_d is the heat transfer coefficient between the workpiece and workbench, ε is the materials emission and k_b is the Stefan-Boltzmann constant.

3. Materials and methods

3.1. Experimental characterization

3.1.1. Preparation of samples

The casting assembly used in solidification experiments consists of water-cooled mold with heat being extracted only from the bottom. It promotes a vertical upward directional solidification and this directional solidification apparatus was used to obtain an Al-1.5 wt% Fe alloy cylindrical casting, with 60 mm diameter and 100 mm length. This alloy was prepared with pure raw materials. The laser surface treatment was performed with a 2 kW Yb-fiber laser (IPG YLR-2000S). The laser beam was focused by a 160 mm lens on sample surface, while the laser wavelength was λ = 1.06 μm and the its intensity at initial moment was $I(0)=1.81\times10^9$ $W\cdot m^{-2}$. The power density was of 4.8×10^5 $W\cdot cm^{-2}$ with multiphase distribution of energy with an approximately Gaussian profile. For this experiment, the sample was positioned 3 mm above the laser focus (out-focusing), using a laser beam diameter of about 600 μm. This laser treatment without an assisting gas jet was applied to propitiate the formation of alumina and nitride on laser-treated surface and to promote the formation of a passive oxide layer in contact with the environment and these phases present high hardness, as discussed by Pariona et al. [9].

3.1.2. Characterization techniques

For the metallographic characterization of cross-section, small samples were cut and sanded with 600, 800, 1200 grit SiC sand paper, and polished with colloidal silica in a semi-automatic polishing machine (AROTEC Ind. e Com., Brazil). Micrographs were recorded by an optical microscopy (OM, Olympus-BX51) and by a scanning electron microscopy (SEM, Shimadzu SSX-550 microscope). Atomic force microscopy analysis was performed in a Shimadzu SPM-9600 microscope with 400-nm minimum resolution, equipped with a 125-μm scanner operating in non-contact mode, according to Pariona et al. [9].

3.2. Numerical implementations

The coupled governing equations were numerically solved employing the commercial software COMSOL [8], optimized by Multigrid technique and the thermophysical properties of Al-1.5 wt.%Fe alloy were assumed to be temperature-dependent, as

Table 1
Thermophysical properties of Al-1.5 wt.%Fe alloy.

Temperature (K)	Density (kg/m^3)	Temperature (K)	Heat capacity (J/kg·K)	Temperature (K)	Thermal conductivity (W/m·K)
934	2620	935	1275	573	140
935	2582.4	936	1275	623	136
936	2519.4	936.5	3333	673	131
937	2491.7	937	109,000	724	129
938	2465.5	937.5	139,000	773	125
940	2450	938	129,000	823	122
		939	81,000	873	118
		940	51,000		
		941	1163		
		942	1163		
		943	1163		
		944	1163		

shown in Table 1, These properties were found through Thermo-Calc software [10].

For simulation process were considerate a tetragonal geometry with: width = 10 mm (x-axis), depth = 4 mm (y-axis) and height = 2 mm (z-axis), the number of tetrahedral elements = 30039 and trouble coupling laminar flow and heat transfer in fluids by Comsol software [8].

The value of $\gamma(T)$ for the Marangoni effect, it represents the temperature derivative of the surface tension was considered as $\gamma(T) = (0.914–3.5 \times 10^{-4}\ (T\text{-}T_{sol}))\ N{\cdot}m^{-1}$ and where liquid–solid phase change occurs at T_{sol} = 933 K for pure aluminum, according to Velde et al. [11], though, in line with Table 1, T_{sol} = 937 K.

Calculations of coupled velocity and temperature field were performed according to Semi-Implicit Method and was optimized by Geometric Multigrid technical, whose parameters used were: number of iterations, 2; Multigrid cycle, V-cycle; hierarchy generation method; lower element order first number of multigrid levels, 1 and mesh coarsening factor, 2.

Numerical approach used in discretization is based on finite element method to solve two dimensional Navier-Stokes equations, assuming incompressible flow. A laminar model and mathematical modeling were provided for solution convergence, nevertheless, a comparison of microstructure with the result of fluid mechanics for laminar case and heat transfer in liquid, involving the Maran-

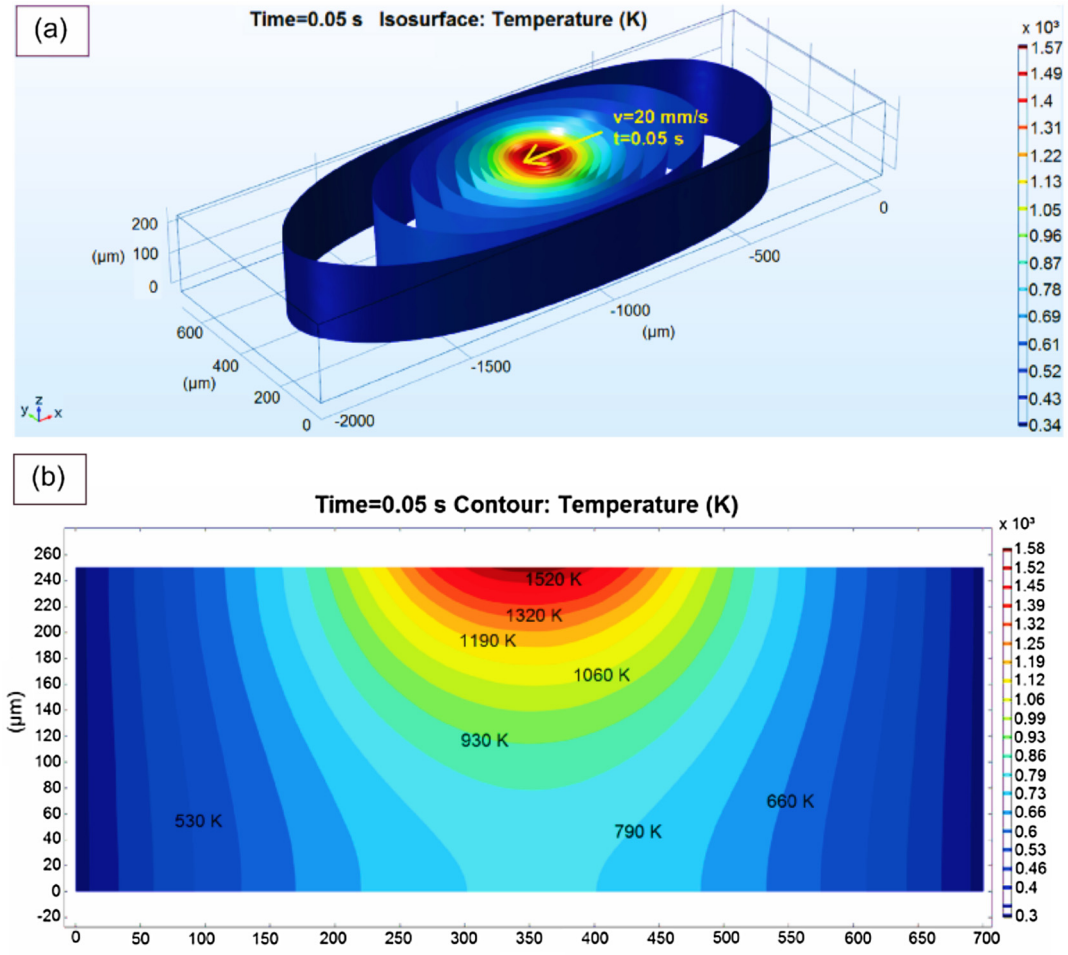


Fig. 1. For v = 20 mm/s. (a) 3-D perspective views of the temperature field at moment t = 0.5 s and (b) isothermal field in the workpiece cross-section.

goni effect, it will be done. For numerical simulation process, same size of micrograph's molten pool geometry was adopted for each velocity of the laser beam.

4. Result and discussion

In this chapter, results of numerical simulation, micrographs and morphology will be presented, discussed and they will be gauged. A 3D thermal FEM model has been built to calculate the temperature variation and fluid flow field in the molten pool cross-section. Although, numerical simulation offers tool of effective evaluation of thermal behavior during laser surface remelting (LSR) process. Simulation was conducted by commercial COMSOL package.

In this section will be discussed influence of the Marangoni effect on microstructure of laser-treated workpiece through presented mathematical modeling. Where, results of simulation were verified with the experimental part.

4.1. The Marangoni effect on microstructure for the laser beam velocity 20 mm/s

4.1.1. Temperature field

As result of material heating, a molten pool was created in the laser beam operation area. Fig. 1a shown the temperature field in 3D on the workpiece surface, where the isotherms are visualized when the laser beam moves at a velocity of 20 mm/s at instant 0.5 s. The temperature field of the molten pool whose geometric characteristic has an ellipse form. This fact by Liu et al. [12] and Hong-yun et al. [13] were confirmed. Beside that, the authors Yang et al. [6] reported that the elliptical weld molten pool and its temperature field were approximately bilateral symmetry relative to the stringer centerline. In the workpiece cross-section (Fig. 1b), also the temperature field is shown, again the isotherms are displayed, to illustrate the temperature field, in different points were exhibited values of the temperatures.

4.1.2. Microstructure in the workpiece cross-section

The micrograph for single track, for the velocity 20 mm/s of the laser beam is shown at Fig. 2, where clearly is noted the molten pool zone and workpiece untreated (Fig. 2a). The geometry of the molten pool showed off a protuberance on this surface, as agreed upon by Bertelli et al. [14], this is due to effect of laser velocity. These authors found that this effect may occur at the laser velocities below 100 mm/s, still, in this zone, nano porous are present. On the other hand in Fig. 2, the width and depth of the molten pool were 680 μm and 257 μm respectively. Besides, on the workpiece untreated revealed a microstructure of Al_3Fe intermetallic phase (white region) dispersed in a matrix phase (gray region), this was supported by Pariona et al. [15]. While at the molten zone, the microstructure containing very fine particles with cellular and enlarged appearance (Fig. 2b), however, this microstructure is constituted of metastable phases, according to Pariona [16]. In Fig. 2c the interface between workpiece untreated and molten pool was magnified for more details and in Fig. 2d this zone was magnified more still, it is worth mentioning here, change of this interface

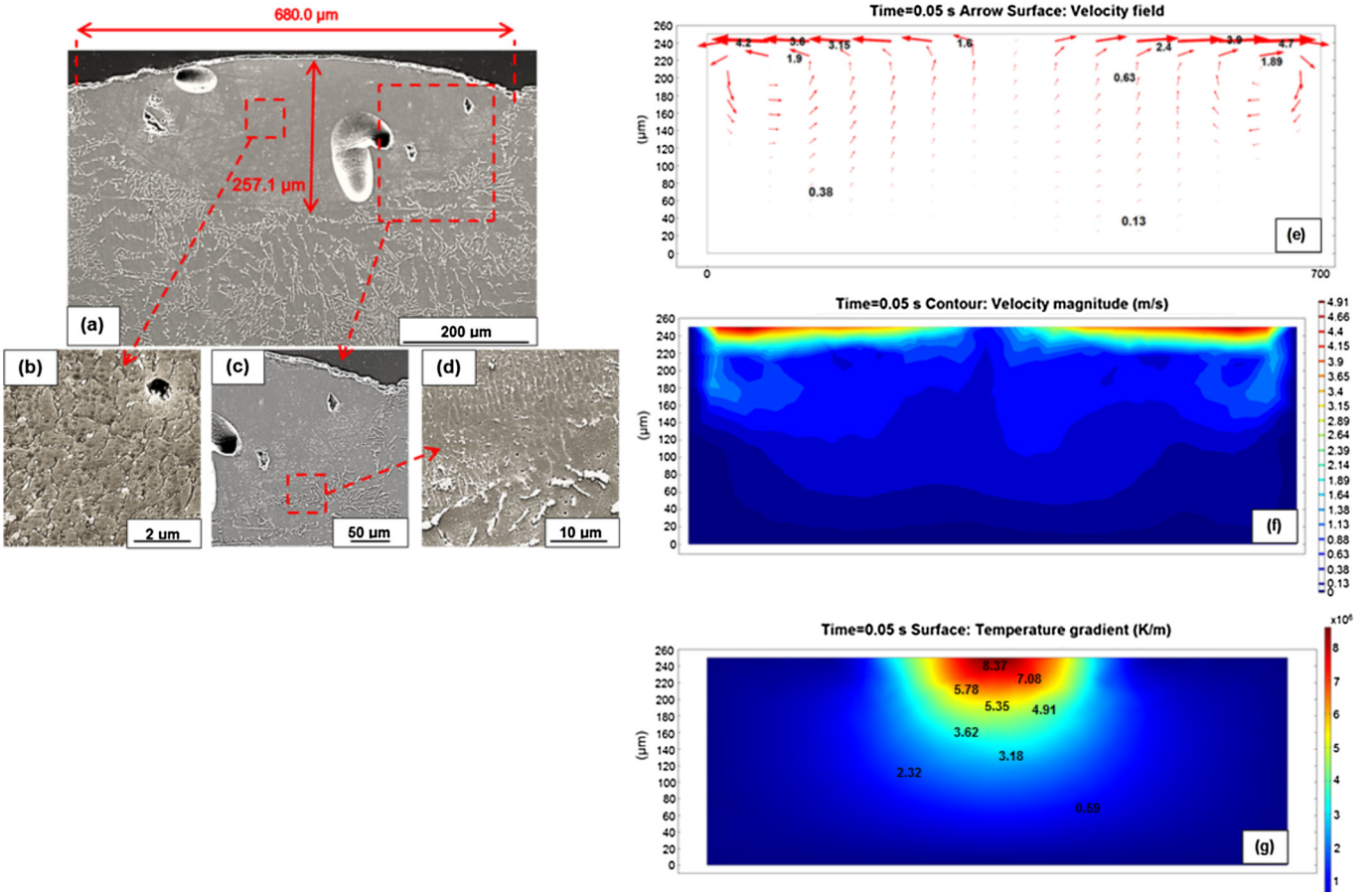


Fig. 2. Shows SEM micrograph in the workpiece cross-section and FEM results of the Marangoni effect of LSR-treated workpiece (v = 20 mm/s): (a) workpiece untreated and molten pool, (b) magnification of the molten pool zone, (c) magnifications of the molten pool-substrate interface, (d) more magnification of the transition zone, (e) fluid-velocity vector field; (f) fluid-velocity field, and (g) thermal gradient.

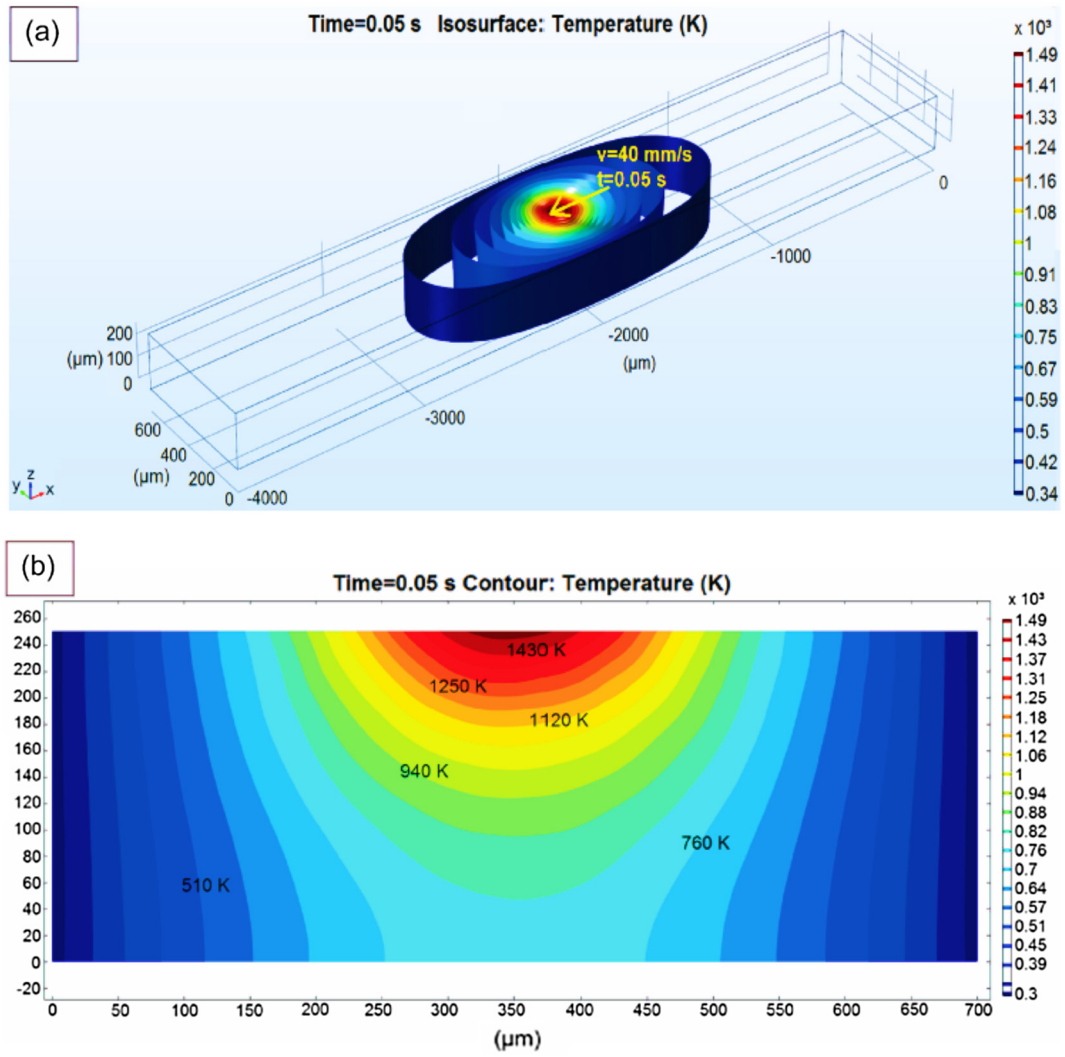


Fig. 3. (a) Computed temperature distribution in 3-D and (b) Isothermal field in the workpiece cross-section.

region type is a little perceptible, Pariona et al. [16] confirmed. How can it be verified, in this region, the treated and untreated zone show their own characteristics at same time.

4.1.3. Velocity field and thermal gradient

Result for the laser beam velocity of 20 mm/s can be seen in Fig. 2e, where the fluid-velocity vector field is shown and in Fig. 2f the fluid-velocity field is observed, all this was, at instant 0.5 s. However, at same figures, the fluid flow presents counter-clockwise (on the left) and clockwise (on the right) vortexes at the molten pool, beside there are two symmetric vortexes corresponding to the molten pool center line. The symmetric vortices can be attributed, due to that the laser beam has Gaussian behavior and the molten pool center line corresponds to the origin of Gaussian geometry, nevertheless qualitatively, in this figure, maximum flow velocity occurs nearby on workpiece surface, however, the fluid flow velocity decreases when the distance moved away from workpiece surface to the molten pool-substrate interface. Quantitatively, the larger velocity was about 5 m/s close of the molten pool surface and the smaller velocity was below 0.13 m/s nearby of the melt pool-substrate interface. This result fits very well with study carried out by Yang et al. [6], who affirmed, the larger velocity indicated strong heat transfer and convection happened in the molten pool area, and the driving force of fluid flow was mainly the surface tension gradient. Marangoni convection made the liquid metal flow from the higher temperature area to the lower temperature area.

The thermal gradient is displayed in Fig. 2g, where, the most prominent values are in vicinity on the workpiece surface and the lower values are found around of the molten pool-substrate interface. Apart from, the maximum of the thermal gradient is around 8.5×10^6 K/m on the workpiece surface and the minimum is around 0.6×10^6 K/m near at the molten pool-substrate interface

4.2. The Marangoni effect on microstructure for the laser beam velocity 40 mm/s

4.2.1. Temperature field

On the other hand, result for the velocity 40 mm/s of the laser beam is displayed at Fig. 3, where in Fig. 3a is visualized the temperature field in 3D on the workpiece surface at instant 0.5 s and Fig. 3b represents the temperature in the cross-section, were the isotherms are displayed.

4.2.2. Microstructure in the workpiece cross-section

Besides, the microstructure for same laser beam velocity in Fig. 4 is displayed. The characteristic of the geometry of the molten

pool, thus, its width and depth were 675 μm and 224 μm respectively. Where, in Fig. 4a an overall view is displayed, in this case, the protuberance was not observed. So that the molten pool zone was magnified (Fig. 4b), thus, the microstructure presented very fine particles with enlarged appearance, slightly different from Fig. 2b, however, the magnification of the molten pool-substrate interface is exhibited in Fig. 4c, wherein behavior is similar to Fig. 2c, nevertheless, its higher magnification (Fig. 4d) is different from Fig. 2d, nevertheless, few elongated rifling of the second phase are visualized in Fig. 4d.

4.2.3. Velocity field and thermal gradient

The fluid-velocity vector field and fluid-velocity field are shown in Fig. 4e and f, respectively, behavior of these fields were similar to the previous result. Whose larger velocity was about 4 m/s on the workpiece surface and the minimum velocity was below 0.1 m/s at the molten pool-substrate interface. The thermal gradient is visualized in Fig. 4g, however, the larger ones are presented in the vicinity on the workpiece surface and the smaller values are found near of the molten pool-substrate interface.

The thermal field magnitude for the velocity of 40 mm/s is less than for the velocity 20 mm/s, as can be confirmed by Figs. 1 and 3. As well as, the thermal gradient was lower for this last velocity (40 mm/s). This can be explained by time that the laser beam stays on workpiece surface, so that, high laser velocities stay less time than for low velocity. Moreover, the maximum of the thermal gradient is around 7.4×10^6 K/m on the workpiece surface and the minimum is around 0.7×10^6 K/m near at the molten pool-substrate interface, therefore, high laser velocities produce smaller size of the molten pool zone, which can be verified in Figs. 2 and 4 and also the Marangoni effect is smaller.

4.3. The Marangoni effect on microstructure for the laser beam velocity 60 mm/s

4.3.1. Temperature field

However, for 60 mm/s of the laser beam velocity, temperature field pattern on the workpiece surface is displayed in Fig. 5a, in the workpiece cross-section is exhibited in Fig. 5b and thus its result were similar for the velocity of 40 mm/s.

4.3.2. Microstructure in the cross-section

Furthermore, result for the microstructure and the Marangoni effect is portrayed in Fig. 6. For this situation, the width and depth were 673 μm and 190 μm respectively, related to geometry of the molten pool. Also, the protuberance was not observed at the microstructure (Fig. 6a), when the molten pool zone was magnified (Fig. 6b), microstructure characteristic is totally different than in low velocities of laser beam. Furthermore, it display many nano porosities and it does not show a well-defined structure, otherwise, magnification of the molten pool-substrate interface is revealed in Fig. 6c, details are totally different than previous cases, nevertheless, when this zone is the highly enlarged (Fig. 6d), it shows a very fine striation, moreover, the second fine phase is observed as if it is emerging from the sides.

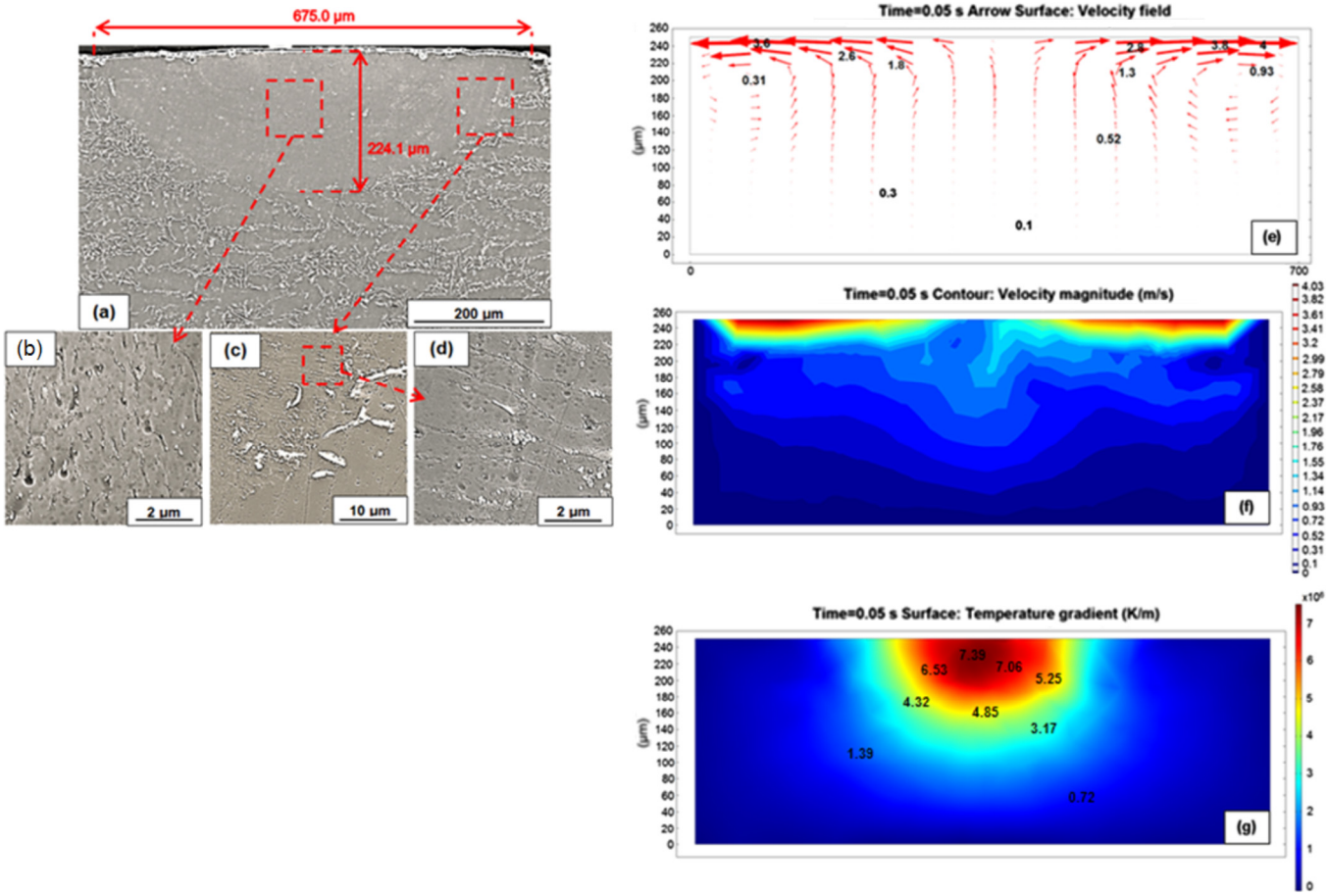


Fig. 4. Microstructure morphology by SEM in the cross-section and FEM results of the Marangoni effect of LSR-treated workpiece (40 mm/s): (a) an overall view, (b) magnification of molten pool zone, (c) magnifications of the molten pool-substrate interface; and (d) high magnification of the transition zone, (e) fluid-velocity vector field; (f) fluid-velocity field, and (g) thermal gradient.

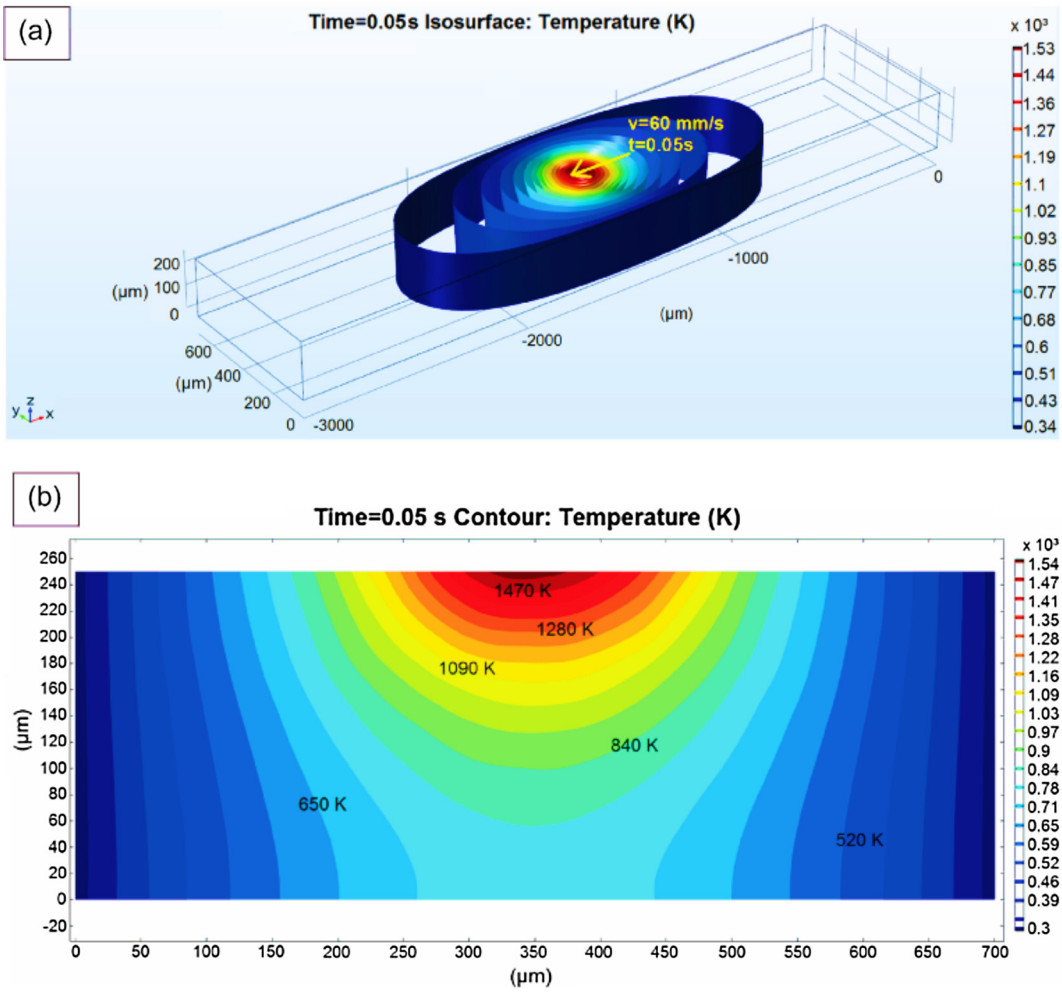


Fig. 5. For v = 60 mm/s. (a) Temperature field on the workpiece surface in 3-D and (b) shows the calculated temperature field evolution in the workpiece cross-section.

4.3.3. Velocity field and thermal gradient

Moreover, in Fig. 6e and f are displayed result of the fluid-velocity vector field and fluid-velocity field, respectively. Where the maximum velocity on the workpiece surface was close to 3.7 m/s, but nearby of the interface was of order 0.09 m/s.

Fig. 6g shows result of the thermal gradient when LSR technique is applied, in this figure we can visualize a mapping of the thermal gradient in the molten pool zone. The maximum of the thermal gradient is around 5×10^6 K/m on the workpiece surface and the minimum is around 1×10^6 K/m near at the molten pool-substrate interface. Meanwhile, the size of geometry of the molten pool, the magnitude of the velocity field and thermal gradient were smaller than for previous velocities of the laser beam. Therefore, the Marangoni effect was lower than for high speeds of the laser beam.

Validation of the simulated geometry of the molten pool with the fusion region from the experiments. As is known, liquid–solid phase change occurs at Tsol = 933 K for pure aluminum, according to Velde et al. [11], though, in line with Table 1, Tsol = 937 K. It should be emphasized; the liquid–solid phase change temperature was calculated based on the phase diagram in near-equilibrium. Nevertheless, LSR-treated process occurs out of equilibrium, in rapid solidification mode and including being able to have a high supercooling temperature, meanwhile, the cooling rate and thermal gradient are too high, yet, high fluid velocity of molten liquid metal occurs due to presence of the Marangoni effect, as a consequence of these phenomena, the grain size is very fine, e.g., for the laser beam velocity of 20 mm/s (Fig. 2(b)), however, for high laser beam velocities the microstructure does not show a well-defined, as already discussed. Nonetheless, for the liquid–solid phase change temperature at 933 K, for this was calculated the depths according to simulation result, so that, by Figs. 1(b), 3 (b) and 5(b), the depths was approximately on average at 130 μm, it presented very small variation for different velocities, meanwhile, this result cannot be used for experimental validation, as explained anteriorly.

In addition, some interesting information can be check in this work, to carry out the validation of the simulation and experimental, so, for simulation results of the Marangoni effect, e.g., in Figs. 2 (f), 4(f) and 6(f), where the simulated depth of the molten pool was approximately 220 μm, for the molten pool-substrate interface region, being that this value varied slightly for different velocities. On the other hand, the depth of experimental part were: 257.1 μm for Fig. 2(a), 224 μm for Fig. 4(a) and 189 μm for Fig. 6(a), respectively. Because the depth of the molten pool decreases as the laser beam velocity increases, although, the mean experimental depth was 223 μm and the simulated mean depth was 220 μm, consequently, both results are close, thus, the simulated geometry of the molten pool is quite similar to the fusion region from the experiments.

Owing to the heights fluid flow velocity and thermal gradient near on the workpiece surface, grain size on microstructure at this zone is smaller than close of the molten pool-substrate interface, for example, in Fig. 2 or Fig. 4 or Fig. 6. Another reason, for exam-

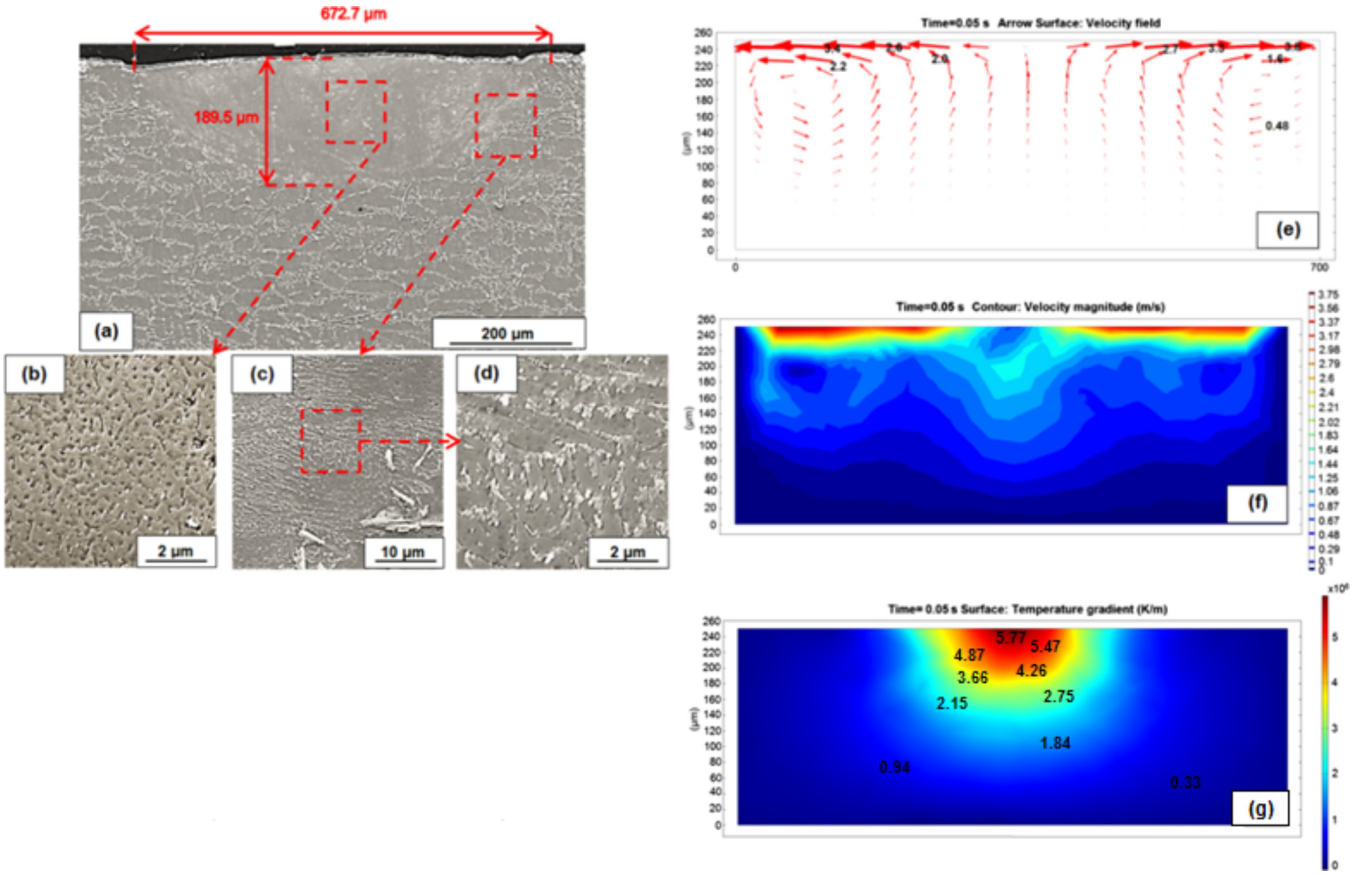


Fig. 6. Microstructure in the cross-section; SEM and FEM results of the Marangoni effect in LSR-treated alloy (60 mm/s): (a) an overall view, (b) magnification of the molten pool zone, (c) magnifications of the molten pool-substrate interface, (d) high magnification of transition zone, (e) fluid-velocity vector field; (f) fluid-velocity field, and (g) thermal gradient.

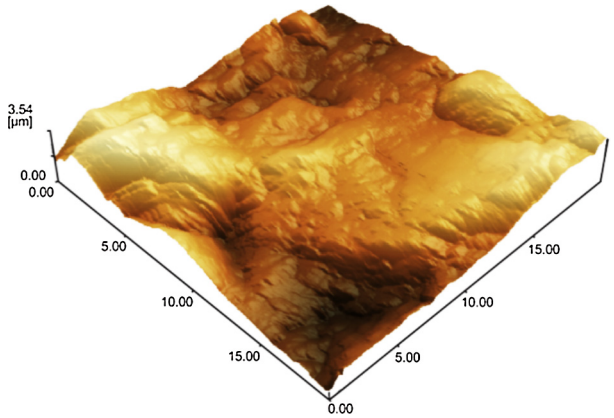


Fig. 7. AFM topography of workpiece untreated without the Maragoni effect.

ple, in Fig. 1 or Fig. 3 or Fig. 5, in the molten pool a high temperature is prominent, especially on the workpiece surface, consequently a high cooling rate is produced and a high thermal gradient is created in this zone (Fig. 2g or Fig. 4g or Fig. 6g), due to this motive the microstructure is thinner around on the workpiece surface, nevertheless, microstructure is composed of metastable phases, especially alumina and aluminum nitride, however, experience was carried out in medium ambient, as it were studied by Pariona et al. [16] and Pariona et al. [17]. Also these same authors have demonstrated that microhardness is higher in treated area than untreated sample, as agreed upon by Pariona et al. [15], this fact proves that the microstructure is very thin and composed especially by alumina and aluminum nitride in laser-treated zone. With this result, it is demonstrated that the Marangoni effect is directly proportional to the fluid flow velocity and thermal gradient and, hence, this affects that controls the quality and properties of microstructure.

4.4. Influence of the Marangoni effect on the laser-treated workpiece morphology

The description and discussion of this section is only introduced based on the experimental results. A notorious influence that the Marangoni effect has on microstructural characteristic of alloy under study was shown, however, also this effect influences morphology characteristic of laser-treated workpiece. In the following, result of influence of the Marangoni effect will be shown on morphology through SEM and atomic force microscopy (AFM). AFM technique providing details of surface on nanometric scale, according to Pariona et al. [18]. The topography of workpiece untreated by AFM image in Fig. 7 shows a non-uniform surface with peaks and valleys, where, the average roughness (Ra) found was about 607.5 μm.

Fig. 8a displays SEM image of as-received laser-treated workpiece for v = 20 mm/s of the laser beam, it reveals numerous defects such as pores scattered throughout the track contour and in intermediate area of track were not observed defects appreciable. However, the topography of laser-treated workpiece by AFM in Fig. 8b displays, a non-uniform surface with peaks and valleys, for this situation, the average roughness (Ra) was around of

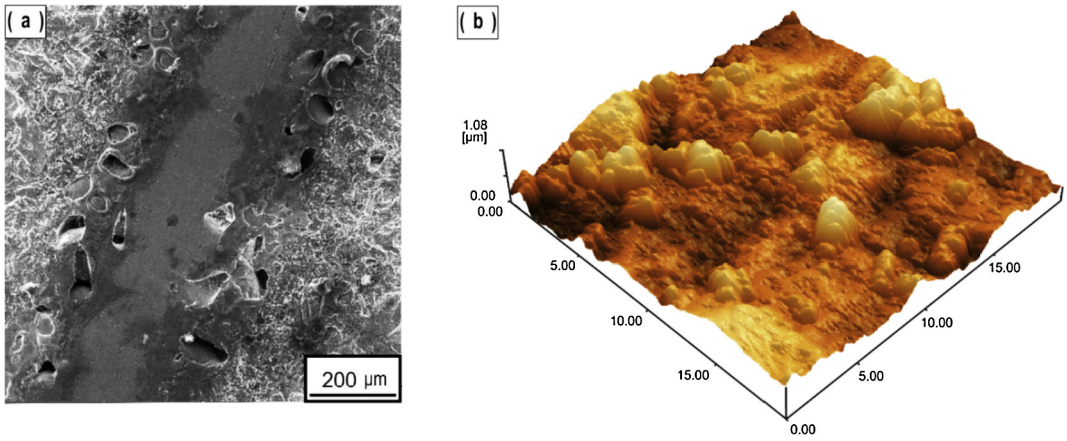


Fig. 8. For laser beam velocity v = 20 mm/s and single track. (a) SEM micrographs of laser-treated workpiece and (b) 3-D perspective views of AFM morphology at track middle.

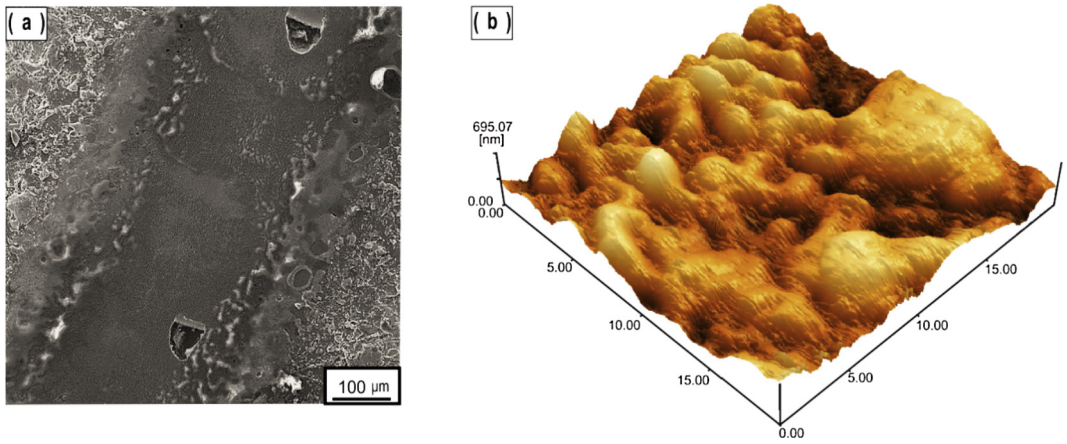


Fig. 9. For v = 40 mm/s of laser beam velocity and single track. (a) Microstructure of laser-treated workpiece by SEM and (b) Surface topography by AFM at track middle.

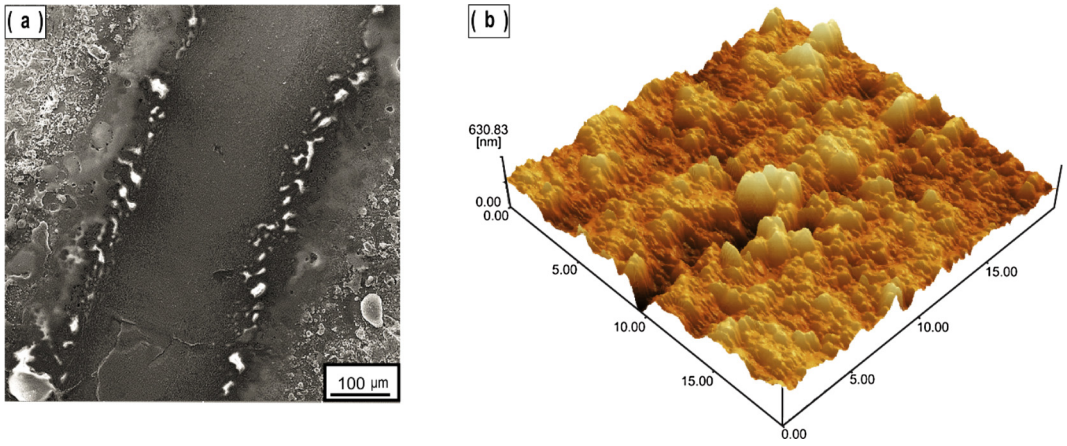


Fig. 10. For v = 60 mm/s of laser beam velocity and single track. (a) SEM exhibition of laser-treated workpiece microstructure and (b) AFM mapping pictures of morphology superficial at track middle.

132.3 µm. Nevertheless, peaks and valleys were much smaller compared to the workpiece untreated.

Fig. 9a illustrates result of SEM microstructure of as-received laser-treated workpiece for the laser beam velocity 40 mm/s, the track reveals few defects, being that this truck was more defined than for v = 20 mm/s, however, it displays some porosities within of track. Furthermore, the morphology of laser-treated workpiece by AFM in Fig. 9b is displays, also a non-uniform surface with peaks and valleys is noted, for this situation, the average roughness (Ra) found was 94.4 µm, nevertheless, peaks and valleys were much smaller compared to the workpiece untreated and laser-treated workpiece (v = 20 mm/s).

In addition, for v = 60 mm/s, SEM microstructure is appreciated in Fig. 10a, where few defects are noticed and the truck has uniform behavior, the morphology (Fig. 10b) for this case is more uniform and lower than for previous results, whose Ra was 61.4 μm.

As can be shown that the Marangoni effect also influences morphological characteristic and as well as quality of laser-treated workpiece track. Therefore, at low laser beam velocities the morphology is higher and quality of track presents many defects than at high laser beam velocities.

5. Conclusions

According to this study that was carried out, the following conclusions will be presented.

(1) In the molten pool a high temperature is prominent, especially on the workpiece surface, consequently a high cooling rate and thermal gradient are produced in this zone, meanwhile, the microstructure is thinner close on the workpiece surface,
(2) In low laser beam velocities, the microstructure presented very fine particles with cellular and enlarged appearance; otherwise, for high velocities it does not show a well-defined microstructure.
(3) Geometry characteristic magnitude of the molten pool decreases as the laser beam velocity increases. The protuberance in the molten pool was not observed in high velocities.
(4) The simulated geometry of the molten pool is quite similar to the fusion region from the experiments.
(5) The Marangoni effect was more noticeable in lower than in high laser beam velocities. It influences and controls quality and properties of microstructure.
(6) The Marangoni effect is directly proportional to the fluid flow velocity and thermal gradient and, hence, this affects that controls the quality and properties of microstructure.
(7) The Marangoni effect influences morphological characteristic, as well as quality of laser-treated workpiece track, therefore, at low laser beam velocities the morphology is higher and quality of track presents many defects than at high laser beam velocities.
(8) Numerical simulation is a powerful tool to obtain complete understanding of physical phenomena and underlying mechanisms of LSR-treated processes, therefore, to better use this innovative technology, for thorough physical understanding associated to molten pool phenomena.

Conflict of interest

The authors declare that there is no conflict of interest.

Acknowledgments

This work was entirely financed by CNPq (Brazilian National Council for Scientific and Technological Development), Fundação Araucária (FA), CAPES (Federal Agency for the Support and Evaluation of Postgraduate Education), and FINEP (Research and Projects Financing Agency). We also thank to LABMU–UEPG.

References

[1] A. Siwek, Numerical simulation of the laser welding, Comput. Sci. 9 (2008).
[2] T.L. Bergman, J.R. Keller, Combined buoyancy, surface tension flow in liquid metals, Numer. Heat Transf. 13(1) (1988).
[3] Z.-S. Mao, J. Chen, Numerical simulation of the Marangoni effect on mass transfer to single slowly moving drops in the liquid–liquid system, Chem. Eng. Sci.. 59 (8–9) (2004) 1815–1828.
[4] Z. Gan, G. Yu, X. He, S. Li, Numerical simulation of termal behavior and multicomponente mass transfer indirect laser deposition of Co-base aloy on steel, Int. J. Heat Mass Transf. 104 (2017) 28–38.
[5] A.B. Murphy, A perspective on arc welding research: the importance of thearc, unresolved questions and future directions, Plasma Chem. Plasma Process. 35 (3) (2015) 471–489.
[6] Z. Yang, W. Tao, L. Li, Y. Chen, C. Shi, Numerical simulation of heat transfer and fluid flow during double-sided laser beam welding of T-joints for aluminum aircraft fuselage panels, Opt. Laser Technol. 91 (2017) 120–129.
[7] X.S. Gao, C.S. Wu, S.F. Goecke, H. Kügler, Numerical simulation of temperature field, fluid flow and weld bead formation in oscillating single mode laser-GMA hybrid welding, J. Mater. Process. Technol. 242 (2017) 147–159.
[8] COMSOL Release notes 5.2, Friction Stir Welding, Stockholm, Sweden, 2016.
[9] M.M. Pariona, F. de Oliveira, V. Teleginski, S. Machado, M.A.V. Pinto, Simulation and optimization of Al-Fe aerospace alloy processed by laser surface remelting using geometric multigrid solver and experimental validation, Heat Mass Transf. 52 (2016) 1037–1049.
[10] THERMO CALC software, Stockholm, Sweden, 2015.
[11] O. Velde, R. Gritzki, R. Grundmann, Numerical investigations of Lorentz force influenced Marangoni convection relevant to aluminum surface alloying, Int. J. Heat Massa Transf. 44 (2001) 2751–2762.
[12] Z. Liu, H. Qi, Mathematical modeling of crystal growth and microstructure formation in multi-layer and multi-track laser powder deposition of single-crystal superalloy, Physics Procedia 56 (2014) 411–420.
[13] Z. Hong-yun, Z. Hong-tao, X. Chun-hua, Y. Xian-qun, Z. Hong-yun, Z. Hong-tao, X. Chun-hua, Y. Xian-qun, Temperature and stress fields of multi-track laser cladding, Trns. Nonferrous Met. Soc. China, 19 (2009) 495–501.
[14] F. Bertelli, E.S. Meza, P.R. Goulart, N. Cheung, R. Riva, A. Garcia, Laser remelting of Al–1.5wt.% Fe alloy surfaces: numerical and experimental analyses, Opt. Lasers Eng. 49 (2011) 490–497, https://doi.org/10.1016/j.optlaseng.2011.01.007.
[15] M.M. Pariona, V. Teleginski, K. Santos, S. Machado, A.J. Zara, N.K. Zurba, R. Riva, Yb-fiber laser beam effects on the surface modification of Al–Fe aerospace alloy obtaining fillet weld structures, low fine porosity and corrosion resistance, Surf. Coat. Technol. 206 (2012) 2293–2301.
[16] M.M. Pariona, V. Teleginski, K. dos Santos, A.A.O.C. de Lima, A.J. Zara, T.M. Micene, R. Riva, Influence of laser surface treated on the characterization and corrosion behavior of Al–Fe aerospace alloys, Appl. Surf. Sci. 276 (2013) 76–85, https://doi.org/10.1016/j.apsusc.2013.03.025.
[17] M.M. Pariona, K.T. Micene, The alumina film morphology formed to improve the corrosion resistance of Al-2.0wt%Fe alloy as result of the laser surface melting technique applied, Adv. Chem. Eng. Sci. 7 (2017) 10–22.
[18] M.M. Pariona, V. Teleginski, K. dos Santos, E.L.R. dos Santos, A.A.O.C. de Lima, R. Riva, AFM study of the effects of laser surface remelting on the morphology of Al–Fe aerospace alloys. Mater. Characteriz. 74 (2012) 4–7.